Contrastive learning for rare genetic disease classification

Submitted by

Nafiz Reza Talukder	**20210204034**
Abu Dojana	**20210204039**
Kazi Zannatul Tajrin	**20210204042**

Supervised by

Professor Dr. S.M.A. Al-Mamun
Professor

Department of Computer Science and Engineering
Ahsanullah University of Science and Technology

Dhaka, Bangladesh

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ABSTRACT

Classifying rare genetic diseases is difficult due to scarce labeled data and fragmented clinical information. To address this, we propose a self-supervised contrastive learning framework that learns gene–phenotype representations without relying on large annotations. We built a dataset by integrating gene metadata and phenotypes from NCBI, DisGeNET, and HPO. Using SimCLR with NT-Xent loss, the model learns biological similarities by comparing augmented views of the same gene. On a cohort of 25 genes, the method achieved a Silhouette Score of 0.287 and formed two meaningful clusters, including a high-severity group linked to developmental delays. Clustering was partly dominated by the common "Autosomal recessive inheritance" phenotype (60%), showing the effect of dataset homogeneity. These results provide a proof-of-concept. Future work will scale to 300 genes to reduce dominant features and enable finer disease subtype separation.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Introduction to the Problem

In the rapidly evolving landscape of bioinformatics and precision medicine, Machine Learning (ML) has emerged as a pivotal tool for deciphering complex biological data. While traditional ML models have achieved remarkable success in diagnosing common diseases where vast amounts of labeled data are available, they often falter when applied to rare genetic diseases. Rare diseases, by definition, have a low prevalence, resulting in a severe scarcity of data for individual conditions. This phenomenon creates a "long tail" distribution where the majority of diseases have very few diagnosed cases, making standard supervised learning techniques inefficient or prone to overfitting.

A significant compounding factor in this domain is the fragmentation of data. Information regarding gene characteristics, phenotypes, and disease associations is often scattered across disparate repositories, making it difficult to train unified models. The core challenge addressed in this thesis is the classification of these rare genetic disorders by overcoming both data scarcity and data fragmentation.

To tackle this, we propose the application of Contrastive Learning, a paradigm that focuses on learning representations by comparing similar and dissimilar pairs of data points. We utilize a custom-curated dataset that integrates gene metadata (such as location and exon count) with phenotypic ontology terms. By leveraging the structure of these relationships—pulling genes with similar phenotypic profiles closer together in the embedding space—we aim to build a robust model capable of classifying rare genes even when training samples are scarce. This research explores how such representation learning can bridge the gap between fragmented, low-volume data and accurate medical diagnosis.

1.2 Motivation

There are estimated to be over 7,000 distinct rare diseases affecting hundreds of millions of people globally. For patients suffering from these conditions, the path to a correct diagnosis is often a "diagnostic odyssey"—a journey involving years of misdiagnoses and ineffective treatments. A primary bottleneck is the lack of a centralized, holistic view of the data; vital information is often siloed in separate databases like NCBI [1], HPO [2], or DisGeNET [3].

From a computational perspective, standard Deep Learning models are "data hungry" and struggle to find decision boundaries with only a handful of positive examples. In the context of rare diseases, finding even fifty confirmed cases for a specific gene mutation can be impossible. This discrepancy motivates the need for a learning paradigm that is "data efficient."

We are motivated by the hypothesis that constructing a comprehensive dataset—combining structural gene attributes (e.g., Chromosome, Exon Count) with deep phenotypic descriptions (HPO terms)—will allow a Contrastive Learning model to capture subtle patterns that standard classifiers miss. By focusing on the intrinsic similarities between these rich feature sets rather than just categorical labels, the model can learn useful features from smaller datasets. The potential to shorten the diagnostic odyssey by synthesizing scattered data into a predictive tool serves as the primary driving force behind this research.

1.3 Objectives

The primary objective of this thesis is to develop a machine learning framework capable of accurately classifying rare genetic diseases by utilizing contrastive learning on a custom-integrated dataset. To achieve this, we have outlined the following specific goals:

- **Custom Dataset Construction:** To aggregate and harmonize data from multiple authoritative sources—specifically the National Center for Biotechnology Information (NCBI) [1], DisGeNET [3], and the Human Phenotype Ontology (HPO) [2]—creating a unified dataset that links gene metadata with their associated HPO terms.
- **Feature Engineering:** To preprocess and encode heterogeneous features, including categorical data (e.g., Chromosome, Gene Type) and variable-length phenotype lists, into a format suitable for neural network input.
- **Implementation of Contrastive Learning:** To design and implement a neural network architecture that utilizes contrastive loss functions to learn meaningful representations based on phenotypic similarity rather than supervised class attributes.

- **Model Optimization:** To experiment with different contrastive strategies to determine the most effective method for separating distinct rare disorders in the latent space.
- **Performance Evaluation:** To evaluate the model's classification accuracy and precision, specifically testing its performance on unseen rare disease data and quantifying the improvements gained over traditional baseline methods.

Chapter 2

Literature Review

The classification of rare genetic disorders presents unique challenges in bioinformatics due to the "long-tail" distribution of disease prevalence and the fragmentation of biological data. This review examines the evolution of Machine Learning (ML) strategies from standard supervised approaches to Self-Supervised Learning (SSL), specifically Contrastive Learning (CL), and their application in medical diagnostics, genomic analysis, and rare disease classification.

2.1 The Shift to Contrastive Self-Supervised Learning

Traditional supervised learning relies heavily on large-scale, annotated datasets. However, in the medical domain, and particularly for rare diseases, obtaining high-quality labeled data is often cost-prohibitive or impossible due to patient scarcity.

Jaiswal et al. [4] provide a comprehensive survey of Contrastive Self-Supervised Learning, highlighting its ability to learn robust representations without explicit labels. The core principle involves pulling "positive" pairs (e.g., augmented views of the same sample) closer in the embedding space while pushing "negative" pairs apart. This is typically achieved using the InfoNCE (Information Noise-Contrastive Estimation) loss function, which maximizes the mutual information between augmented views. Architectures such as SimCLR, MoCo, and BYOL have demonstrated that models trained on pretext tasks (such as rotation prediction or color jittering) can learn invariant features that generalize remarkably well to downstream tasks. This paradigm is particularly relevant to our research, as it offers a mathematical framework to handle class imbalance by focusing on feature distinguishability rather than class probability alone.

2.2 Contrastive Learning in Medical Imaging

The majority of current research applying Contrastive Learning to healthcare focuses on Medical Imaging (MRI, PET, Dermoscopy). These studies validate the efficacy of CL in extracting diagnostic features from complex, noisy biological data.

Alzheimer's Disease (AD) Classification: Several studies have successfully utilized CL for neurodegenerative disorders. Fiasam et al. [5] proposed Domain Contrastive Learning (DCL) to address the variability in MRI data across different clinical sites. By aligning feature distributions from multiple domains, they improved model transferability. Similarly, Chen et al. [6] applied contrastive strategies to 18F-FDG PET imaging. They addressed the issue of small sample sizes and low signal-to-noise ratios in PET scans by generating extended training data through cropping and adopting a contrastive loss that minimizes intra-class differences while maximizing inter-class distances.

Dermatological Classification: In the domain of skin disease, Aboulmira et al. [7] demonstrated that Contrastive Learning significantly enhances classification performance compared to traditional Cross-Entropy methods. Their work highlighted the model's ability to differentiate between visually similar skin lesions by learning fine-grained feature representations, a crucial capability when dealing with diseases that share overlapping phenotypic characteristics.

2.3 Multimodal and Genomic-Aware Architectures

While imaging-based CL is well-established, integrating genomic and non-image data remains a developing frontier.

Multimodal Integration (ContIG): Taleb et al. [8] introduced ContIG, a framework for Self-Supervised Multimodal Contrastive Learning. This method aligns medical images with genetic data in a shared feature space. By integrating six genetic modalities (including SNPs and CNVs) with retinal images, ContIG demonstrated that leveraging the interplay between phenotype (image) and genotype (genetics) enhances performance on downstream tasks. However, this approach still relies heavily on the availability of dense imaging data, which is often unavailable for rare genetic disorders defined primarily by clinical ontology terms.

Gene Networks and Ontology (CoGO): Closer to the text-based nature of genetic classification, Chen et al. [9] proposed CoGO, a contrastive framework for predicting disease similarity. CoGO utilizes Gene Ontology (GO) graphs and gene interaction networks to learn low-dimensional representations of diseases. By employing a graph-based contrastive approach, they successfully identified relationships between diseases based on molecular mechanisms. While CoGO validates the use of graph structures in CL, its primary focus is on measuring *similarity* rather than the direct *classification* of disorders based on ragged phenotypic inputs.

2.4 Challenges in Rare Disease Classification

The application of Machine Learning to rare genetic diseases has been extensively studied, yet significant hurdles remain. Roman-Naranjo et al. [10] conducted a systematic review of ML approaches in this domain, concluding that while techniques like Random Forests and Support Vector Machines (SVMs) have improved diagnostic yields, they often struggle with the high dimensionality of genomic data and the phenotypic heterogeneity of rare conditions. Their review emphasizes that "one-size-fits-all" models are insufficient and highlights the critical need for integrative models that can handle diverse and fragmented biological data sources.

This classification challenge is further complicated by data privacy and scarcity. Siganporia et al. [11] explored Privacy-Enhanced Federated Learning for rare genetic disorders using Electronic Health Records (EHR). Their work highlights the "data scarcity" issue, where individual hospitals lack sufficient cases to train robust models. They utilized a federated approach where local model updates are aggregated to a global model without exchanging raw patient data. While this addresses privacy, they rely on standard supervised classification heads which may struggle with the extreme class imbalance inherent to rare diseases.

2.5 Research Gap

The literature establishes that Contrastive Learning is a powerful tool for handling unlabeled data and class imbalance. However, a distinct gap exists:

1. **Modality Gap:** Most medical CL research focuses on dense pixel data (Images) or continuous signals. There is limited research on applying contrastive learning to sparse, categorical lists of HPO (Human Phenotype Ontology) terms and Exon counts.
2. **Application Gap:** While ContIG integrates genetics with images, and CoGO looks at disease similarity, there is a lack of unified frameworks that directly classify rare genetic disorders by combining Gene Metadata and Phenotypic Ontologies.

This thesis addresses these gaps by developing a Contrastive Learning framework specifically tailored for non-image, variable-length biological data to improve the diagnosis of rare genetic disorders.

Chapter 3

Proposed System

3.1 Workflow Diagram

The proposed system follows a sequential pipeline designed to transform fragmented, raw biological data into meaningful disease clusters. The workflow is divided into five critical stages as shown in Figure 3.1:

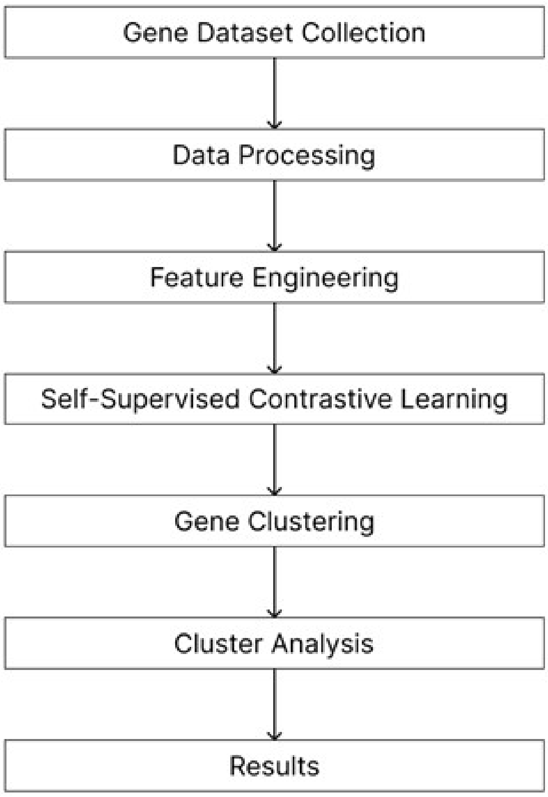

Figure 3.1: System workflow showing the five critical stages from data input to analysis

Phase 1: Data Input & Preparation Aggregates raw phenotype and disease data into a unified dataset while correcting missing gene symbols using forward-fill logic.

Phase 2: Feature Engineering Transforms clinical symptoms into multi-hot binary vectors and encodes biological metadata to create comprehensive, hybrid feature vectors for each gene.

Phase 3: Self-Supervised Training Trains a neural network using contrastive loss to pull augmented views of the same gene together and push different genes apart, without using disease labels.

Phase 4: Embedding & Clustering Extracts learned gene embeddings from the encoder and groups them into distinct disease families using hierarchical clustering with an optimal cluster count.

Phase 5: Output & Analysis Visualizes the resulting clusters using dimensionality reduction techniques and interprets them biologically to identify meaningful disease categories.

3.2 Chosen Machine Learning Method

The core methodology chosen for this research is Self-Supervised Contrastive Learning, specifically adapted for genomic data using the SimCLR (Simple Framework for Contrastive Learning of Visual Representations) paradigm.

Unlike traditional supervised learning, which requires explicit labels for every sample (often unavailable for rare diseases), this method learns by comparison. The model does not initially know which gene causes which disease. Instead, it learns a "disease space" through the following mechanism:

- **Positive Pairs:** Two artificially altered views of the *same* gene (e.g., one with missing symptoms, one with noise added). The model is trained to recognize these as representations of the same biological entity.
- **Negative Pairs:** Views of *different* genes. The model is trained to distinguish these from the positive pair.

By forcing the model to pull positive pairs together and push negative pairs apart in the embedding space, the system learns to identify the intrinsic features that define a gene's pathological profile.

3.3 System Performance Factors

The success of this system depends on three key factors:

1. **Hybrid Feature Engineering:** We combine high-dimensional phenotype data with low-dimensional biological metadata (chromosome location, exon count) to create a comprehensive representation of each gene.
2. **Augmentation Robustness:** Since the dataset is small, the efficacy of the augmentation strategy (phenotype masking and noise injection) is crucial to ensure the model never sees the exact same input twice.
3. **Loss Function Suitability:** We use the NT-Xent (Normalized Temperature-scaled Cross Entropy) Loss, which is mathematically designed for contrastive learning scenarios, making it superior to standard classification loss for this task.

Chapter 4

Dataset Analysis

4.1 Source and Credibility of Datasets

Unlike standard machine learning domains where pre-compiled datasets are readily available, the field of rare genetic diseases suffers from severe data fragmentation. No single repository currently contains the comprehensive mapping of gene attributes, phenotypic associations, and disease classifications required for this research. Consequently, we constructed a custom dataset by aggregating and harmonizing data from three authoritative biological databases:

1. **National Center for Biotechnology Information (NCBI)** [1]: This served as the primary source for official gene nomenclature, chromosomal locations, and structural attributes (such as exon counts).

2. **DisGeNET** [3]: Used in conjunction with NCBI to verify gene-disease associations and obtain official gene symbols.

3. **Human Phenotype Ontology (HPO) / hpo.jax.org** [2]: The authoritative source for standardized phenotypic terminology. This database provides a controlled vocabulary for describing observable clinical features (phenotypes).

By cross-referencing these globally recognized databases, we ensured the credibility and medical accuracy of the data. The final dataset integrates structural biological data with clinical phenotypic descriptions, creating a unified resource suitable for contrastive learning.

4.2 Dataset Statistics and Description

The dataset consists of 10 distinct attributes per record, categorized into Gene Identification, Structural Attributes, and Clinical Associations.

Current Data Volume:

As of this reporting phase, the pilot dataset includes 25 distinct genes serving as the initial seed for our model. These genes are linked to a rich set of clinical features, comprising 100 distinct disease entities and 2043 total HPO associations (of which approximately 1100 are unique phenotypic terms). The final target for the completed thesis is to scale this dataset to include 300 distinct genes, providing a broader coverage of the rare disease landscape.

Feature Description:

Below is a description of each feature included in our analysis:

Table 4.1: Dataset Feature Description

Attribute Name	**Category**	**Description**
Official Full Name of Gene	Gene ID	The complete, biologically standardized name of the gene (e.g., *BCL2 apoptosis regulator*).
Official Symbol of Gene	Gene ID	A unique, short-form alphanumeric identifier (e.g., *BCL2*). This symbol links data across databases.
Gene Type	Structural	Classifies the gene based on function (e.g., "protein-coding," "ncRNA").
Location	Structural	The specific cytogenetic location on the chromosome (e.g., *18q21.33*).
Exon Count	Structural	A numerical value indicating the number of coding regions in the gene structure.
Chromosome	Structural	Indicates the chromosome number (1-22, X, Y) where the gene resides.
HPO Association	Clinical	The descriptive name of the phenotype (e.g., *Abnormal peritoneum morphology*).
HPO Term Identifier	Clinical	The unique ID assigned by the Human Phenotype Ontology [2] (e.g., *HP:0002585*).
Disease Name	Clinical	The specific rare genetic disorder linked to the gene (e.g., *Follicular lymphoma*).
Disease Identifier	Clinical	The unique accession number (e.g., *ORPHA:545*) identifying the disease entity.

4.3 Anomalies and Challenges in the Dataset

During the data aggregation and inspection process, we identified specific structural imbalances and anomalies. Addressing these is critical for the robustness of our Contrastive Learning model.

1. Variable Density of Phenotypic Associations (Ragged Data)

The most significant anomaly observed is the lack of uniformity in the number of associations per gene.

- **Observation:** The depth of clinical annotation varies wildly between genes. For instance, a well-documented gene in our dataset may have over 50 distinct phenotypic associations, while another gene might be linked to only a single specific disorder.
- **Impact:** This creates a "ragged" data structure where input vectors are of variable length. Standard neural networks typically require fixed-size inputs, necessitating a sophisticated encoding strategy to handle these variable feature sets.

2. Class Imbalance (Phenotypic Rarity)

There is a distinct imbalance in the frequency of specific phenotypic features.

- **Observation:** Common phenotypes (e.g., terms related to "Kidney" abnormalities) appear in roughly 100 samples, whereas rarer, more specific phenotypes might appear in fewer than 5 samples.
- **Impact:** This imbalance risks biasing the model towards common symptoms while ignoring rare but diagnostic features. The contrastive learning approach helps mitigate this by focusing on relative similarity rather than absolute frequency.

3. Data Cleaning and Integrity

Due to the automated nature of merging data from different sources, some records initially contained empty or impossible values (e.g., missing Exon Counts).

- **Handling:** To maintain integrity, we performed rigorous data cleaning. We skipped records with impossible or empty values during the scraping process.
- Consequently, the final dataset contains only fully populated, verifiable biological instances.

Official Full Name	Official Symbol	Gene type	Location	Exon Count	Chromosome	HPO Term Identifier	HPO Association
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0002585	Abnormal peritoneum morphology
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001541	Ascites
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0012378	Fatigue
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001945	Fever
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0003072	Hypercalcemia
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0025435	Increased circulating lactate dehydrogenase concentration
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0002716	Lymphadenopathy
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001004	Lymphedema
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0002665	Lymphoma
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0100721	Mediastinal lymphadenopathy
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0033823	Mediastinal mass
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001287	Meningitis
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0030166	Night sweats
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0002202	Pleural effusion
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0200036	Skin nodule
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001744	Splenomegaly
BCL2 apoptosis regulator	BCL2	protein coding	18q21.33	4	18	HP:0001824	Weight loss

Figure 4.1: HPO Association Dataset Sample

Official Full Name	Official Symbol	Disease Identifier	Disease Name
BCL2 apoptosis regulator	BCL2	ORPHA:545	Follicular lymphoma
checkpoint kinase 2	CHEK2	ORPHA:668	Osteosarcoma
checkpoint kinase 2	CHEK2	OMIM:259500	Osteosarcoma
checkpoint kinase 2	CHEK2	OMIM: 176807	Prostate cancer
checkpoint kinase 2	CHEK2	OMIM:609265	Li-Fraumeni syndrome 2
checkpoint kinase 2	CHEK2	ORPHA: 145	Hereditary breast and/or ovarian cancer syndrome
checkpoint kinase 2	CHEK2	ORPHA:440437	Familial colorectal cancer Type X

Figure 4.2: Disease Association Dataset Sample

Chapter 5

System Implementation

5.1 Data Preprocessing

The raw data from NCBI [1], DisGeNET [3], and HPO [2] was processed to create a unified input tensor.

- **Gap Filling:** We apply "Forward Fill" logic to handle missing gene symbols in the source CSVs, correcting artifacts where Excel merged cells left gaps.
- **Matrix Creation:** We convert the list of 2,000+ phenotype associations into a **Multi-hot Encoded Matrix**. For roughly 1,300 unique phenotype terms, each gene gets a binary vector where '1' indicates the presence of a symptom and '0' its absence.
- **Normalization:** We scale all features using a StandardScaler. This is crucial for normalizing continuous variables (like Exon Count) so they are on the same scale as the binary phenotype features.

5.2 Data Augmentation Strategy

To simulate a larger dataset and enforce robust learning, we implemented a custom GeneAugmentation class. Every time a gene is fed into the model, it undergoes stochastic transformations:

1. **Phenotype Masking:** We randomly mask (set to 0) approximately 15% of the active symptoms. This mimics real-world clinical scenarios where patient data is incomplete.

2. **Noise Injection:** We add Gaussian noise ($\mu = 0$, $\sigma = 0.1$) to the continuous biological features (e.g., Exon Count). This prevents the model from memorizing exact values and forces it to learn relative distributions.

5.3 Model Architecture

We designed a custom Encoder-Projection Neural Network in PyTorch:

1. **The Encoder:** A Multi-Layer Perceptron (MLP) that compresses the high-dimensional input ($\sim$1,321 features) into a compact latent representation (64 dimensions). It uses ReLU activations, Batch Normalization after each layer to stabilize training.
2. **The Projection Head:** A temporary small network ($64 \rightarrow 32$) used exclusively during training to project embeddings into the space where contrastive loss is calculated.

5.4 Training Process

The model is trained using a self-supervised loop for 100 epochs with the Adam optimizer (Learning Rate = 0.001) and NT-Xent Loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

This loss function calculates the probability that sample i and sample j are a positive pair, normalized by the similarity of sample i to all other samples in the batch.

Data Splitting Strategy: To ensure rigorous evaluation, we split the dataset into three distinct subsets:

- **Training Set (70%):** Used to update the model weights. The model sees augmented views of these genes to learn representations.
- **Validation Set (15%):** Used to tune hyperparameters and monitor for overfitting during training. The model does not learn from this data but is evaluated on it at the end of every epoch.
- **Test Set (15%):** A completely unseen set of genes held out until the very end to evaluate the final clustering performance.

5.5 Embedding Extraction

Upon completion of training, the "Projection Head" is discarded. We pass the entire dataset through the frozen "Encoder" network to extract the Gene Embeddings. These embeddings are 64-dimensional vectors that serve as the "fingerprint" for each gene. In this latent space, genes that cause similar diseases are mathematically closer to each other than those that do not.

5.6 Clustering Analysis

We employ unsupervised clustering on the extracted embeddings to discover disease groups:

- **Algorithm:** We use Agglomerative Hierarchical Clustering with Ward's linkage. This method constructs a tree of clusters, which is more interpretable for biological data than other clustering methods like K-Means.
- **Optimal K Selection:** The system automatically determines the optimal number of clusters by testing a range (K=2 to K=8) and selecting the value that maximizes the Silhouette Score, ensuring the clusters are distinct and cohesive.

5.7 Performance Metrics and Analysis

We evaluated the Self-Supervised Contrastive Learning model using standard clustering validity indices. Given the small sample size (25 genes), the results indicate a moderate but meaningful separation of disease types.

- **Silhouette Score (0.287):** This score falls within the "moderate" range (-1 to +1). It indicates that the model successfully identified clusters that are more similar within groups than between groups, despite the complex, shared symptoms of rare diseases.
- **Davies-Bouldin Index (1.747):** This "fair" score suggests the clusters are mathematically distinguishable but not perfectly compact.
- **Training Convergence:** The training loss decreased steadily from 2.52 to 1.02 over 100 epochs, confirming that the model successfully learned meaningful representations from the data augmentation used.

5.8 Cluster Interpretation

The model automatically grouped the 25 genes into 2 distinct clusters based on phenotypic similarity.

Cluster 0: Developmental & Structural Disorders (4 Genes)

- **Genes:** *DOCK8, G6PC3, SALL1, SDHD*
- **Key Insight:** This is a high-severity cluster. 100% of these genes are associated with "Failure to thrive," and 75% with "Global developmental delay." This cluster effectively separates severe, early-onset systemic conditions from the rest.

Cluster 1: Neurological & Systemic Disorders (21 Genes)

- **Genes:** *BCL2, CHEK2, UBE3A, SCN10A, etc.*
- **Key Insight:** This broad cluster is dominated by neurological symptoms, with 52% of genes linked to "Seizures" and 42% to "Intellectual disability." The large size suggests these genes share common pathways or are grouped due to shared generic symptoms.

5.9 Results Visualization

Three key visualizations confirmed the model's performance:

1. **t-SNE Scatter Plot (Figure 5.1):** The 2D projection visually separates the small, tight Cluster 0 from the larger, more dispersed Cluster 1, confirming the Silhouette analysis.
2. **Training Loss Curve (Figure 5.2):** The curve shows rapid learning in the first 20 epochs and stabilization by epoch 80, proving the training duration was efficient.
3. **Hierarchical Dendrogram (Figure 5.3):** The tree diagram clearly splits into two primary branches at the top level, validating our choice of $K = 2$ as the most natural division for this dataset.

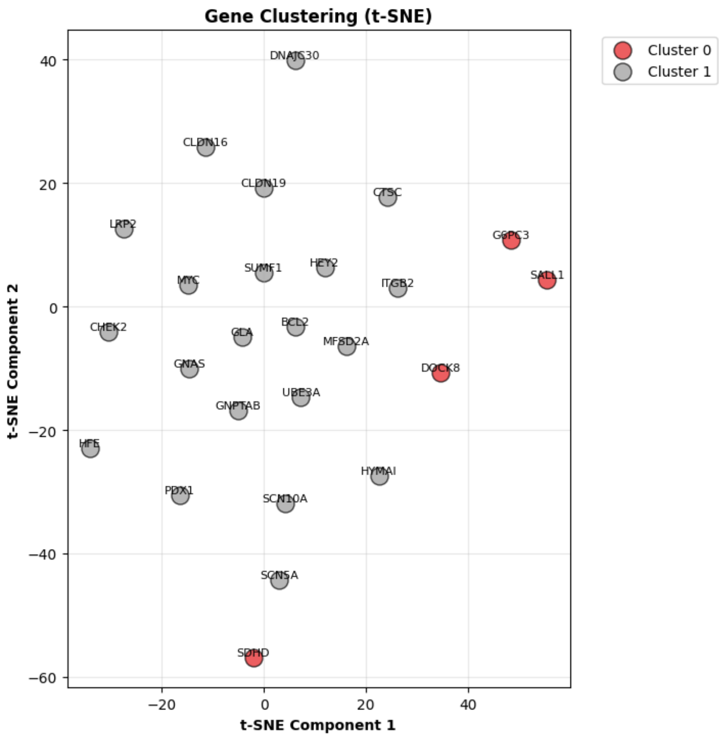

Figure 5.1: t-SNE visualization of gene clusters showing clear separation between developmental disorders (Cluster 0) and neurological/systemic disorders (Cluster 1)

Figure 5.2: Training loss convergence over 100 epochs showing successful model learning

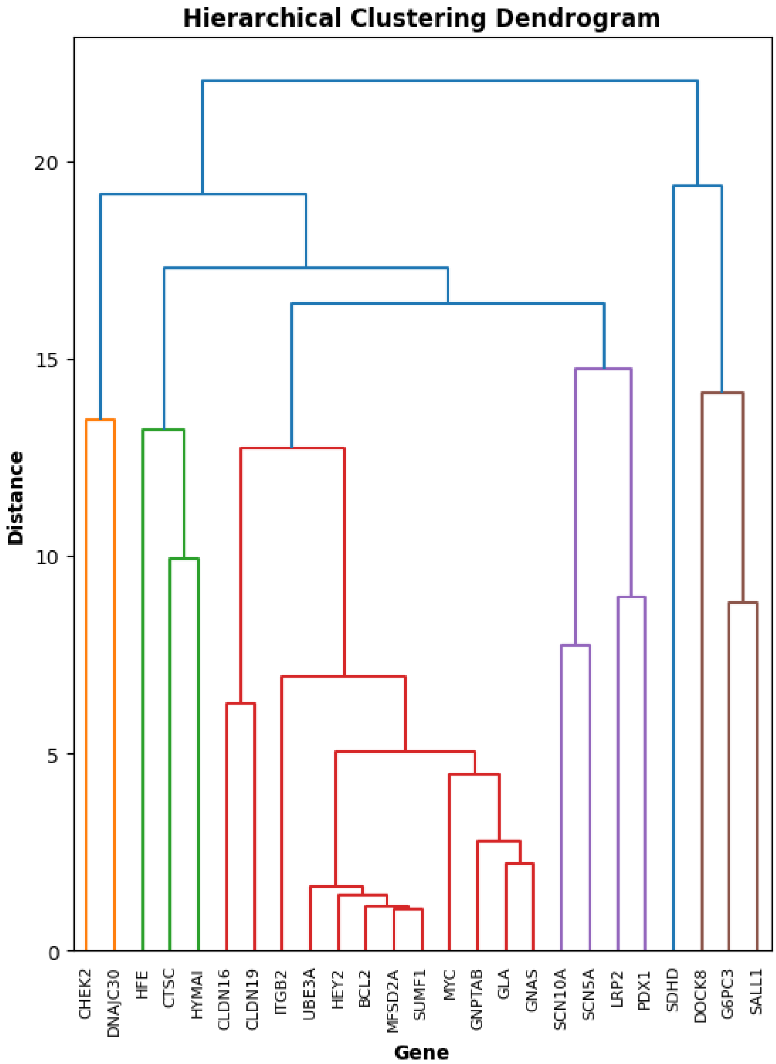

Figure 5.3: Hierarchical clustering dendrogram showing the natural division of genes into two main clusters

Chapter 6

Observation and Conclusion

6.1 Discussion

The experimental results confirm that our Self-Supervised Contrastive Learning model successfully extracted meaningful disease structures from a small, unlabeled dataset of 25 genes. With a Silhouette Score of 0.287, the model identified 2 distinct clusters purely based on phenotypic similarity.

A major success was the isolation of Cluster 0, which grouped genes linked to severe, early-onset conditions like "Failure to thrive" and "Global developmental delay." This proves the model can detect strong biological signals without manual supervision.

However, Cluster 1 emerged as a broad, mixed group containing 21 of the 25 genes. Analysis revealed the reason: most genes are clustered under one disease type because most genes of the dataset contain a common phenotype: Autosomal recessive inheritance (15 genes, 60.0% of all genes). This shared feature dominated the learning process, causing the model to group diverse conditions (like cancers and neurological disorders) together rather than separating them into finer subtypes.

6.2 Conclusion and Future Work

This thesis established a successful proof-of-concept for classifying rare genetic diseases using contrastive learning. By leveraging multi-hot phenotype encoding and stochastic augmentation, we demonstrated that meaningful gene embeddings can be learned even from "ragged" and limited data.

While the model effectively separates high-severity developmental disorders from general conditions, the granularity of the clusters is limited by the dataset's small size and homogeneity. Future work will focus on expanding the dataset to hundreds of genes to dilute the dominance of common features like inheritance patterns, allowing the model to learn finer decision boundaries and create a more detailed disease taxonomy.

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